Ideation Phase
Brainstorm & Idea Prioritization

Date	27 October 2026
Team ID	SWUID2026403463
Project Name	HotelPro – Online Hotel Room Booking & Management System
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization

A brainstorming session gives the team room to put every idea on the table before any of them is judged. The focus is on generating many ideas, building on one another's suggestions and only then deciding which ones deserve to be built. For HotelPro, the team used this approach to turn the problems found in the project scope into a ranked list of features and a clear build order.

Reference: https://www.mural.co/templates/brainstorm-and-idea-prioritization

Step-1: Team Gathering, Collaboration and Select the Problem Statement

The team agreed on a few ground rules before starting:

- Defer judgment: no idea is rejected while ideas are being collected.
- Go for quantity first; refine quality afterwards.
- Build on other people's ideas instead of replacing them.
- Stay tied to the problem statements and the project modules.

Selected problem statement (How might we…): How might we let guests find an available room for their dates, see what it looks like and book and pay without conflicts or double bookings, while giving hotel staff and admins simple tools to manage rooms, bookings, check-ins and revenue?

Input	Source
Problems to solve	PS-1 to PS-6 from the Define the Problem Statements document (double bookings on overlapping dates, no live room availability, rooms shown without photos, unclear payment status, manual check-in/check-out records, no admin dashboard)
Scope reference	Project scope: modules for authentication, rooms, availability, booking, payment, check-in/check-out, notifications, admin dashboard and analytics
Team format	Four members, each taking one area: room discovery, booking and payment, hotel operations, admin and future ideas

Step-2: Brainstorm, Idea Listing and Grouping

Each team member silently wrote down five ideas for their area, then shared them with the group. The board below shows all 20 ideas collected.

Person 1: Room discovery	Person 2: Booking and payment	Person 3: Hotel operations	Person 4: Admin and future
#1 Room search by check-in / check-out dates with live availability	#6 Overlap-proof booking with date clash check	#11 Room management with photos and amenities	#16 Admin dashboard and revenue analytics
#2 Filters by room type, price, capacity	#7 Price summary (nights × rate) and guest capacity check	#12 Check-in / check-out management	#17 Cancellation with refund policy

Person 1: Room discovery	Person 2: Booking and payment	Person 3: Hotel operations	Person 4: Admin and future
#3 Room photo gallery with amenities	#8 Coupons and offers	#13 Live occupancy view per date	#18 Email and in-app notifications
#4 Room details page with booked dates	#9 Online payment with status tracking	#14 Housekeeping and maintenance status	#19 AI chatbot for booking help
#5 Personalised room recommendations	#10 My Bookings with printable invoice	#15 Role-based access for guest, staff and admin	#20 Loyalty points and membership

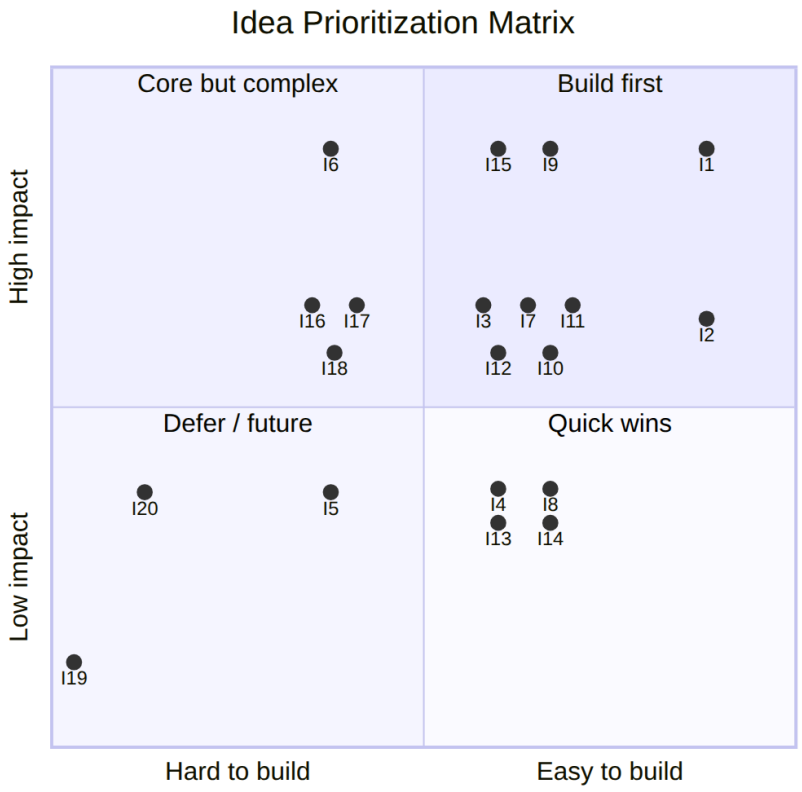
Related ideas were then grouped into six themes:

Theme	Ideas	Count
A. Discovery and search	#1 Room search by dates with live availability; #2 Filters by room type, price, capacity; #3 Room photo gallery with amenities; #4 Room details page with booked dates	4
B. Booking and checkout	#6 Overlap-proof booking; #7 Price summary and guest capacity check; #8 Coupons and offers; #9 Online payment with status tracking	4
C. Bookings and after-booking	#10 My Bookings with printable invoice; #17 Cancellation with refund policy; #18 Email and in-app notifications	3
D. Hotel and room operations	#11 Room management with photos and amenities; #12 Check-in / check-out management; #13 Live occupancy view; #14 Housekeeping and maintenance status; #15 Role-based access	5
E. Admin and insights	#16 Admin dashboard and revenue analytics	1
F. Future innovation	#5 Personalised room recommendations; #19 AI chatbot for booking help; #20 Loyalty points and membership	3

> **Observation:** theme D holds the most ideas, followed by themes A and B, which matches the project scope where room availability, conflict-free booking and room management are the heart of the system.

Step-3: Idea Prioritization

Each idea was scored from 1 to 5 for **impact** (how much it helps guests and the hotel) and **feasibility** (how easy it is to build within the semester using React, Node.js/Express and MongoDB). The score is *impact × feasibility*, and the matrix places each idea in a quadrant.



In the chart, "I" + number = idea ID (e.g. I6 = idea #6). The quadrant boundary sits at 3.5 on both axes, and points are slightly offset where several ideas share the same score. The exact scores are in the table below.

Prioritized idea list (sorted by score):

ID	Idea	Impact	Feasibility	Score	Quadrant	Decision
#1	Room search by dates with live availability	5	5	25	Build first	MVP – Sprint 1
#2	Filters by room type, price, capacity	4	5	20	Build first	MVP – Sprint 1
#15	Role-based access for guest, staff and admin	5	4	20	Build first	MVP – Sprint 1
#9	Online payment with status tracking	5	4	20	Build first	MVP – Sprint 3
#3	Room photo gallery with amenities	4	4	16	Build first	MVP – Sprint 2
#7	Price summary and guest capacity check	4	4	16	Build first	MVP – Sprint 2
#11	Room management with photos and amenities	4	4	16	Build first	MVP – Sprint 2
#12	Check-in / check-out management	4	4	16	Build first	MVP – Sprint 3
#10	My Bookings with printable invoice	4	4	16	Build first	MVP – Sprint 3
#6	Overlap-proof booking with date clash check	5	3	15	Core but complex	MVP – Sprint 2
#16	Admin dashboard and revenue analytics	4	3	12	Core but complex	MVP – Sprint 4

ID	Idea	Impact	Feasibility	Score	Quadrant	Decision
#17	Cancellation with refund policy	4	3	12	Core but complex	MVP – Sprint 4
#18	Email and in-app notifications	4	3	12	Core but complex	MVP – Sprint 4
#4	Room details page with booked dates	3	4	12	Quick win	Include – Sprint 2
#13	Live occupancy view per date	3	4	12	Quick win	Include – Sprint 3
#8	Coupons and offers	3	4	12	Quick win	Include – Sprint 4
#14	Housekeeping and maintenance status	3	4	12	Quick win	Include – Sprint 4
#5	Personalised room recommendations	3	3	9	Defer / future	Basic popularity version – Sprint 4
#20	Loyalty points and membership	3	2	6	Defer / future	Future enhancement
#19	AI chatbot for booking help	2	1	2	Defer / future	Future enhancement

Conclusion

- Ideas in the **Build first** and **Core but complex** quadrants form the minimum viable product: date-based availability search and filters, room photos, conflict-free booking, payment tracking, check-in/check-out, role-based access, cancellations, notifications and the admin dashboard.
- **Quick wins** (room details with booked dates, live occupancy view, coupons, housekeeping status) are added when sprint capacity allows.
- Recommendations start as a basic popularity-based version; the AI chatbot and loyalty programme are recorded as future enhancements.
- Core but complex ideas, especially overlap-proof booking, receive early testing because they carry the highest risk of double booking.